Which one of the following is equivalent to AND-OR realization?

A NAND-NOR realization
B NOR-NOR realization
C NOR-NAND realization
D NAND-NAND realization

Que. 2 The minimal function that can detect a " divisible by 3" 8421 BCD code digit is given by

A D3D1+D4D2+D3D2D1
B D3D1+D4D2D1+D1'D2D1+D3D4D2'D1'
C D4D1+D4D2+D3D1'D2'D1
D D4D2D1'+D4'D2D1+D3D4D2D1

Que. 3 choose the correct statement from the following

A PROM contains a progrmmable AND array and a fixed OR array
B PLA contains a fixed AND array and a programmable OR array
C PROM contains a fixed AND array and a programmable OR array
D PLA contains a programmable AND array and a programmable OR array

Que. 4 A PLA can be used

A as a microprocessor
B as a dynamic memory
C to realise a sequential logic
D to realise a combitional logic

Que. 5 A dynamic RAM consist of

A 6 transistors
B 2 transistors and 2 capacitors
C 1 transistors and 1 capacitors
D 2 capacitors only

Que. 6 When a CPU is interrupted, it

A stops execution of instrution
B acknowledges interrupt and branches of subroutine
C acknowledges interrupt and continues
D acknowledges interrupt and waits for the next instrutions from the interrupting device

Que. 7 The advantage of using a dual slop ADC in a digital voltmeter is that

A its conversion time is small
B its accuracy is high
C its gives output in BCD FORMAT
D it does not required a comparator

Que. 8 The minimum number of comparators required to build an 8 bit flash ACD is

A 8
B 63
C 255
D 256

Que. 9 The number of address lines in a 8 bit 4k ROM is

A 8
B 10
C 12
D 16

Que. 10 The master-slave flip flop has the characteristic that

A change in the input immediately in the output
B change in the output occurs when the state of the master is affected
C change in the output occurs when the state of the slave is affected
D both the master and slave states are affected at the same time

Que. 11 A divide by 78 counter can be realized by using

A 6 nos. of mod-13 counters
B 13 nos. of mod-10 counters

C one mod-13 counters followed by one mod-6 counter
D 13nos. and mod-13 counters
Que. 12 Decimal number 5 in level parity self correcting code
A 00101
B 01011
C 01100
D 10000
Que. 13 The resolution of a D/A convertor is approximately 0.4% of its full scale range.it is
A an 8-bit convertor
B a 10-bit convertor
C a 12-bit convertor
D a 16-bit convertor
Que. 14 The switching speed of ECL is very high,because
A the transistors are switched between cutoff and saturation regions
B the transistors are switched between active and saturation regions
C the transistors are switched between active and cutoff regions
D the transistors may be operated in three regions
Que. 15 _
the logical expression Y=A + AB is equal to
A y=AB
B y=AB
C y=A+B
D y=A+B
Que. 16 Which of the following use least power?
A TTL
B ECL
C CMOS
D all use same power
Que. 17 A system has a word length of 4-bits.If in this system -ve numbers are represented by their Two's compliment,then the range of numbers that can be represented by the word length is
A -8 to +8
B -7 to +7
C -16 to +16
D none of these
Que. 18 Hamming codes are used for error dection and correction.if the minimim hammining distance is m, then the number of errors correctable is
A equal to m
B less than m/2
C equal to 2m
D greater than m
Que. 19 An analog voltage is in the range of 0 to 8v is divided in eight equal intervals for conversion to 3-bit digital output the maximum quantization error is
A 0v
B 0.5v
C 1v
D 2v
Que. 20 Dule slop integration type A/D converters provide
A higher speeds compaired to all other types of A/D converters
B very good accuracy without putting extreme requirments on component stability
C poor rejection of power supply hum
D better resolution compared to all other type of A/D convertors for the same number ofbits

Que. 21 In an 8085 microprocessor system with memory mapped I/O
A I/O devices have 16-bit address
B I/O devices are accessed using IN and OUT instructions
C there can be maximum of 256 input devices and 256 output devices
D arithmetic and logic ooperation can be directly performed with the I/O data.
Que. 22 four memory clips of 16*4 size have their address bases connected together. The system will be of size
A 64*64
B 16*16
C 32*8
D 256*1
Que. 23 A switch tail ring counter is made byusing a single D FF.The resulting ckt is
A SR flip flop
B JK flip flop
C D flip flop
D T flip flop
Que. 24 By placing an inverter between both the input of an SR flip flop,it becomes
A JK flip flop
B D- flip flop
C T flip flop
D master slave JK flip flop
Que. 25 A 12-bit ADC is operating with a 1 micro sec clock period and the total conversion time is seen to be 14micro sec the ADC must be of the
A flash type
B counting type
C integrating type
D successive approximation type
Que. 26 in a 4-bit weighted resistor D/A convertor the resistor value corresponding to LSB is 16kw. the resistor value corrosponding to the MSB will be
A 1Kohm
B 2Kohm
C 4Kohm
D 16Kohm
Que. 27 The number of Boolean functions that can be generated by a variables, is equal to
A 2^n
B 2^2n
C 2^(2n-1)
D 2^n-2
Que. 28 A 0 to 6 counter consist of 3 flip flop and a combinational circuit of 2 input gate(s).
A one AND gate
B one OR gate
C one AND gate and one OR gate
D two AND gates
Que. 29 In order to add 1111+1101, we required
A one FA,one HA
B three FA
C three FA,one HA
D none of these
Que. 30 The signed magnitude representation, the binary equivalent of 2205625 is
A 0,10110.1011
B 0,10110.1001

C 1,10101.1001
D 1,10110.1001

Que. 31 The number of bits needed to encode all letters (26),10 symbols and all numbers (10) is

A 5
B 7
C 6
D 8

Que. 32 The use of a cache in a computer system increases the

A available memory space for the program
B available memory space for data
C avrage speed of memory acces
D addressing range of CPU

Que. 33 A microprocessor has 24 address lines and 32 data lines if it uses 10-bits of op-code the size of its memory buffer register is

A 22 bits
B 24 bits
C 32 bits
D 14 bits

Que. 34 Schottky damping is resosted to in TTL gates

A to reduce propogation delay
B to increases the noise margin
C to increase the packing margin
D to increase fan-out

Que. 35 A pulse train can be delayed by a finite number periods using of clock

A a serial-in-serial out shift resistor
B a serial -in parallel out shift resistor
C a parallel -in series out shift resistor
D a serial -in parallel shift resistor

Que. 36 The switching speed of ECL is very high,because

A the transistors are switched between cutoff and saturation regions
B the transistors are switched between active and saturation regions
C the transistors are switched between active and cutoff regions
D the transistors may be operated in three regions

Que. 37 The number of comparators in a parallel conversion type 8-bit A to D converter is

A 8
B 16
C 255
D 256

Que. 38 A dynamic RAM cell which holds 5v has to be refreshed every 20m secs, so that the stored voltage does not fall by more than 0.5v. if the cell has a constant discharge current of 0.1pA the storage capacitance of the cell is

A 4*10^-6F
B 4*10^-9 F
C 4*10^-12F
D 4*10^-15F

Que. 39 The minimized from of the logical expression (ABC+ABD+ABC+ABC)

A AC+BC+AB

B $\overline{AC}+BC+AB$

C $\overline{A}\,\overline{C}+BC+AB$

D $\overline{AC}\,\overline{B}+BC+AB$

Que. 40 For a binary half-subtractor having two inputs A and B the correct set of logical expression for the output D(=A minus) and X(=barrow) are

A $D=\bar{A}B+A\bar{B}$, $X=\bar{A}B$

B $D=\bar{A}B+A\bar{B}+AB$, $X=\bar{A}B$

C $D=\bar{A}B+A\bar{B}$, $X=\bar{A}B$

D $\overline{D}=\bar{A}B+A\bar{B}$, $X=\bar{A}B$

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